

Lab 10 Tiggers

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MySQL 8.0 Command Line Cli X + v

mysql> CREATE TABLE Student (
    ->     StudentID INT PRIMARY KEY,
    ->     Name VARCHAR(50) NOT NULL,
    ->     Email VARCHAR(100) UNIQUE,
    ->     Course VARCHAR(50) NOT NULL
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql> CREATE TABLE Student_Count (
    ->     Total_Students INT
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql> INSERT INTO Student_Count VALUES (0);
Query OK, 1 row affected (0.01 sec)

mysql> DELIMITER //
mysql>
mysql> CREATE TRIGGER after_student_insert
    -> AFTER INSERT ON Student
    -> FOR EACH ROW
    -> BEGIN
    ->     UPDATE Student_Count
    ->     SET Total_Students = Total_Students + 1;
    -> END //
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> DELIMITER ;
mysql> INSERT INTO Student (Name, Email, Course)
    -> VALUES ('Rahul', 'rahul@college.edu', 'BCA');
ERROR 1364 (HY000): Field 'StudentID' doesn't have a default value
mysql> INSERT INTO Student (StudentID, Name, Email, Course)
    -> VALUES (101, 'Rahul', 'rahul@college.edu', 'BCA');
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO Student (StudentID, Name, Email, Course)
    -> VALUES (102, 'Ruthu', 'ruthu@college.edu', 'BCA');
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO Student (StudentID, Name, Email, Course)
    -> VALUES (103, 'Mrudula', 'mrudula@college.edu', 'BCA');
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mysql> SELECT * FROM Student;
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StudentID	Name	Email	Course
101	Rahul	rahul@college.edu	BCA
102	Ruthu	ruthu@college.edu	BCA
103	Mrudula	mrudula@college.edu	BCA

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3 rows in set (0.01 sec)
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mysql> SELECT * FROM Student_Count;
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Total_Students
3

```
1 row in set (0.00 sec)
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mysql> USE CompanyDB;
Database changed
mysql> CREATE TABLE Employee (
    ->     Emp_ID INT PRIMARY KEY,
    ->     Name VARCHAR(50) NOT NULL,
    ->     Job_Title VARCHAR(50),
    ->     Department VARCHAR(50),
    ->     Email VARCHAR(100) UNIQUE,
    ->     Salary DECIMAL(10,2) NOT NULL,
    ->     Hire_Date DATE DEFAULT (CURRENT_DATE)
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql> CREATE TABLE Salary_Audit (
    ->     Audit_ID INT AUTO_INCREMENT PRIMARY KEY,
    ->     Emp_ID INT,
    ->     Old_Salary DECIMAL(10,2),
    ->     New_Salary DECIMAL(10,2),
    ->     Change_Time TIMESTAMP DEFAULT CURRENT_TIMESTAMP
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql> DELIMITER //
mysql>
mysql> CREATE TRIGGER after_salary_update
    -> AFTER UPDATE ON Employee
    -> FOR EACH ROW
    -> BEGIN
    ->     IF OLD.Salary <> NEW.Salary THEN
    ->         INSERT INTO Salary_Audit (Emp_ID, Old_Salary, New_Salary)
    ->         VALUES (OLD.Emp_ID, OLD.Salary, NEW.Salary);
    ->     END IF;
    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
mysql> INSERT INTO Employee (Emp_ID, Name, Job_Title, Department, Email, Salary)
    -> VALUES (1, 'Ruthu', 'Software Engineer', 'IT', 'ruthu@company.com', 50000.00);
Query OK, 1 row affected (0.01 sec)

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mysql>
mysql> DELIMITER ;
mysql> INSERT INTO Employee (Emp_ID, Name, Job_Title, Department, Email, Salary)
    -> VALUES (1, 'Ruthu', 'Software Engineer', 'IT', 'ruthu@company.com', 50000.00);
Query OK, 1 row affected (0.01 sec)

mysql>
mysql> INSERT INTO Employee (Emp_ID, Name, Job_Title, Department, Email, Salary)
    -> VALUES (2, 'Vaishnavi', 'HR Manager', 'HR', 'viashnavi@company.com', 45000.00);
Query OK, 1 row affected (0.01 sec)

mysql> UPDATE Employee
    -> SET Salary = 60000.00
    -> WHERE Emp_ID = 1;
Query OK, 1 row affected (0.04 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> UPDATE Employee
    -> SET Salary = 50000.00
    -> WHERE Emp_ID = 2;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT * FROM Employee;
+-----+-----+-----+-----+-----+-----+-----+
| Emp_ID | Name   | Job_Title   | Department | Email                      | Salary | Hire_Date |
+-----+-----+-----+-----+-----+-----+-----+
|      1 | Ruthu  | Software Engineer | IT         | ruthu@company.com         | 60000.00 | 2026-09-29 |
|      2 | Vaishnavi | HR Manager   | HR         | viashnavi@company.com     | 50000.00 | 2026-09-29 |
+-----+-----+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)

mysql> SELECT * FROM Salary_Audit;
+-----+-----+-----+-----+-----+
| Audit_ID | Emp_ID | Old_Salary | New_Salary | Change_Time |
+-----+-----+-----+-----+-----+
|      1   |      1 | 50000.00   | 60000.00   | 2026-09-29 12:05:01 |
|      2   |      2 | 45000.00   | 50000.00   | 2026-09-29 12:05:12 |
+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)

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mysql> USE CompanyDB2;
Database changed
mysql> CREATE TABLE Employee (
->     Emp_ID INT PRIMARY KEY,
->     Emp_Name VARCHAR(50) NOT NULL,
->     Department VARCHAR(50),
->     Email VARCHAR(100) UNIQUE NOT NULL,
->     Salary DECIMAL(10,2)
-> );
Query OK, 0 rows affected (0.04 sec)

mysql> CREATE TABLE Deleted_Employee_Log (
->     Log_ID INT AUTO_INCREMENT PRIMARY KEY,
->     Emp_ID INT,
->     Emp_Name VARCHAR(50),
->     Department VARCHAR(50),
->     Email VARCHAR(100),
->     Final_Salary DECIMAL(10,2),
->     Deleted_By VARCHAR(100),
->     Deleted_Time TIMESTAMP DEFAULT CURRENT_TIMESTAMP
-> );
Query OK, 0 rows affected (0.04 sec)

mysql> INSERT INTO Employee VALUES (101, 'Ruthu', 'IT', 'ruthu@company.com', 55000.00);
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO Employee VALUES (102, 'Mrudula', 'HR', 'mrudula@company.com', 48000.00);
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO Employee VALUES (103, 'Vaishnavi', 'Management', 'vaishnavi@company.com', 95000.00);
Query OK, 1 row affected (0.00 sec)

mysql> INSERT INTO Employee VALUES (104, 'Rohith', 'Finance', 'rohith@company.com', 62000.00);
Query OK, 1 row affected (0.00 sec)

mysql> INSERT INTO Employee VALUES (105, 'Sagar', 'IT', 'sagar@company.com', 50000.00);
Query OK, 1 row affected (0.01 sec)

mysql> DELIMITER //
mysql> CREATE TRIGGER before_employee_delete
-> BEFORE DELETE ON Employee

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mysql> DELIMITER //
mysql> CREATE TRIGGER before_employee_delete
-> BEFORE DELETE ON Employee
-> FOR EACH ROW
-> BEGIN
->     IF OLD.Department = 'Management' THEN
->         SIGNAL SQLSTATE '45000'
->         SET MESSAGE_TEXT = 'Restricted Operation: Management employees cannot be deleted from the system.';
->     END IF;
-> END //
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> DELIMITER ;
mysql> DELIMITER //
mysql> CREATE TRIGGER after_employee_delete
-> AFTER DELETE ON Employee
-> FOR EACH ROW
-> BEGIN
->     INSERT INTO Deleted_Employee_Log (Emp_ID, Emp_Name, Department, Email, Final_Salary, Deleted_By)
->     VALUES (OLD.Emp_ID, OLD.Emp_Name, OLD.Department, OLD.Email, OLD.Salary, USER());
-> END //
Query OK, 0 rows affected (0.01 sec)
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```
mysql> DELIMITER ;
mysql> DELETE FROM Employee WHERE Emp_ID = 102;
Query OK, 1 row affected (0.01 sec)
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```
mysql> SELECT * FROM Employee;
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Emp_ID	Emp_Name	Department	Email	Salary
101	Ruthu	IT	ruthu@company.com	55000.00
103	Vaishnavi	Management	vaishnavi@company.com	95000.00
104	Rohith	Finance	rohith@company.com	62000.00
105	Sagar	IT	sagar@company.com	50000.00

4 rows in set (0.00 sec)

```
mysql> SELECT * FROM Deleted_Employee_Log;
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Log_ID	Emp_ID	Emp_Name	Department	Email	Final_Salary	Deleted_By	Deleted_Time
1	102	Mrudula	HR	mrudula@company.com	48000.00	root@localhost	2026-09-29 12:11:34

1 row in set (0.00 sec)